

Tanuj Haresh Sidhwa

Roll No.:230003073

B.Tech

Major: Mechanical Engineering

Minor: Astronomy

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	9.42 (Current)	2023-Present
B.Tech. Minor	Indian Institute of Technology Indore	9.67 (Current)	2024-Present
Higher Secondary	HSC Board	93.67%	2023
Secondary	SSC Board	86.80%	2021

EXPERIENCE

- IASc-INSNA-NASI Summer Research Fellowship**

May 2025 - Jul. 2025

Research Intern / Dr Abhijit Kushari, Advanced Propulsion Laboratory

Indian Institute of Technology, Kanpur

 - Conducted detailed static structural analysis of serpentine nozzles using ANSYS to evaluate deformation behavior under fluid-induced forces and inertial loading encountered in propulsion and high-speed flow systems.
 - Performed fatigue analysis of nozzle under complex loading conditions to estimate lifespan and safety factor, utilizing mean stress correction models such as Goodman, and Gerber using ANSYS to ensure structural reliability.
 - Implemented dynamic analysis using modal and harmonic simulations to characterize structural response, and generated Campbell diagrams to evaluate resonance risks, ensuring integrity across a range of excitation frequencies.
 - Studied plasticity model theory and its characterization; understood implementation in FEM software through bilinear isotropic hardening models, enhancing theoretical and practical understanding of plastic behavior
 - Gained proficiency in CATIA V5 for basic design tasks; conducted preliminary Fluid-Structure Interaction (FSI) studies, supported by a literature review on nozzle deformation using loosely coupled algorithmic modeling methods.

PROJECTS

- Assessment of Indoor Air Quality in Student Hostel**

Sept. 2024 - Oct. 2025

Prof. Ritunesh Kumar, Department of Mechanical Engineering

[GitHub](#)

 - Developed an integrated Arduino-based monitoring system combining NDIR (MH-Z19B) and DHT11 sensors to collect continuous CO₂ concentration and environmental data for indoor air quality assessment in residential spaces.
 - Conducted controlled experiments under varying occupancy levels to monitor CO₂ buildup across different ventilation conditions, capturing time-dependent concentration trends patterns inside student hostel rooms.
 - First author of a paper based on the project, accepted at International Conference on Thermofluids Engineering (INCOTHERM) 2025, IIT(ISM) Dhanbad, Oct. 2025.
- Powered Lift Wing Design for STOL Applications**

Oct. 2025 - Dec. 2025

InterIIT Tech Meet 14.0: High Prep – LAT Aerospace

 - Led a multidisciplinary team to design a high-lift STOL wing, coordinating aerodynamics, propulsion, and stability efforts while managing timelines, task delegation, and system-level integration across workstreams.
 - Drove key technical decisions including airfoil selection, high-lift device configuration, and powered-lift concepts through trade-off studies balancing lift augmentation, drag penalties, stability margins, and computational cost.
 - Supervised and reviewed CFD-driven design iterations in ANSYS Fluent and Workbench, ensuring consistency between 2D, semi-2D, and 3D analyses, and alignment with analytical models.
 - Guided development and validation of multi-fidelity aerodynamic models, integrating theoretical predictions, automated CFD workflows, and parametric studies to achieve Lift Coefficient of 6.81.
 - Consolidated technical findings into a comprehensive end-evaluation report, synthesizing analytical modeling, numerical simulations, and performance insights for competition review and future design scalability.
- CubeSat Project**

Oct. 2024 - Nov. 2024

CAE Club & Astronomy Club, IIT Indore

[GitHub](#)

 - Led the team on the project, overseeing work distribution, workflow coordination, and cross-domain communication for smooth and coherent project progress; also handled official documentation like proposal and report drafting
 - Designed the Primary Chassis of the CubeSat model and conducted detailed modal, thermal, and structural analysis to ensure and validate mechanical integrity, thermal resilience, and stability under expected operational stresses.
 - Executed precise load calculations and verified compliance with aerospace design standards, ensuring the structure could withstand launch conditions and maintain functionality throughout orbital deployment and mission lifespan

• IITI SoC'24

June 2024- Aug. 2024

CubeSat - Quasar 1.0

[GitHub](#)

- Led the team for a 1U CubeSat, overseeing workflow planning, task delegation, inter-domain coordination, and milestone tracking to ensure timely, coherent progress across structural design, analysis, and documentation.
- Oversaw key design decisions including material selection and cost estimation, guiding the team through trade-off analysis to balance performance, manufacturability, and budget constraints suitable for space applications.
- Reviewed and coordinated simulation efforts in ANSYS Workbench, ensuring consistency across structural, thermal, and modal analysis outputs with the competition's design objectives and system-level integration requirements.

• Obstacle Avoidance Line Follower Project

Apr. 2024

Makerspace Lab (IC152), IIT Indore

- Contributed to the hardware design and overall system integration of an obstacle-avoidance line follower robot, with a focus on optimal sensor placement, efficient motor control logic, and reliable power management circuitry.
- Assisted in developing and optimizing circuit layouts focusing on compactness and efficient physical wiring, ensuring reliable performance in real-time obstacle detection and line-following behavior within constrained hardware space.

TECHNICAL SKILLS

- **Analysis:** Ansys: Static and Transient Structural, Modal, Harmonic, Thermal & Fluent, XFLR, XFOIL
- **Design:** Fusion360, CATIA V5, Ansys SpaceClaim
- **Programming:** Python, Numpy*, C++
- **MatLab:** Numerical Methods * Elementary proficiency

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Multivariable Calculus, Differential Equations, Real Analysis, Complex Analysis, Numerical Methods
- **Institute Core:** Basics of Physics, Engineering Mechanics
- **Mechanical Core:** Fluid Machinery, Fluid Mechanics, Strength of Materials, Solid Mechanics, Theory of Manufacturing, Thermodynamics, Principles of Industrial Engineering, Machining Science and Metrology, Heat Transfer, Instrumentation and Control Systems
- **Electives:** Fundamentals of Vibrations, Fundamentals of Acoustics, Introduction to Robotics
- **Minor:** Introduction to Astronomy, Flight mechanics and classical control, IoT for Space Applications

CERTIFICATIONS

- **Rocket Propulsion and Spacecraft Dynamics**, Kodacy, Credential: [5c42207cb6c82268](#) Jul. 30, 2024
- **Aerospace Engineering: Aircraft Fundamentals and Advanced**, Udemy, Credential: [UC-fd19f93f-0e91-4d91-bec0-4d2a76e5941d](#) Oct. 2, 2024

POSITIONS OF RESPONSIBILITY

- **President**, Mechanical Engineering Student Association, IIT Indore Apr. 2025 - Present
- **Core Member**, Astronomy Club, IIT Indore Nov. 2023 - Present
- **Core Member**, CAE Club, IIT Indore Aug. 2024 - Present
- **Project Lead**, CubeSat Project, IIT Indore Oct. 2024 - Nov. 2024
- **Team Lead**, High Prep LAT Aerospace, InterIIT Techmeet 14.0 Oct. 2025 - Dec. 2025

ACHIEVEMENTS

- **Top 1 Percentile in JEE 2023**, Ranked among the top 1% of candidates in JEE 2023, taken by approximately a million students 2023
- **Top 0.1 Percentile in MHT-CET**, Ranked among the top 0.1% of candidates in MHT-CET 2023, taken by approximately 6 Lakh students 2023
- **Reliance Foundation Undergraduate Scholar**, selected for Reliance Foundation Undergraduate Scholarship 2024
- **Bronze Medal:** IITISoC'24, Design & Modelling, Material Selection, Design and FEA of CubeSat 2024
- **6th Position:** InterIIT Techmeet 14.0, Lat Aerospace HighPrep PS, Powered Lift Wing Design 2025